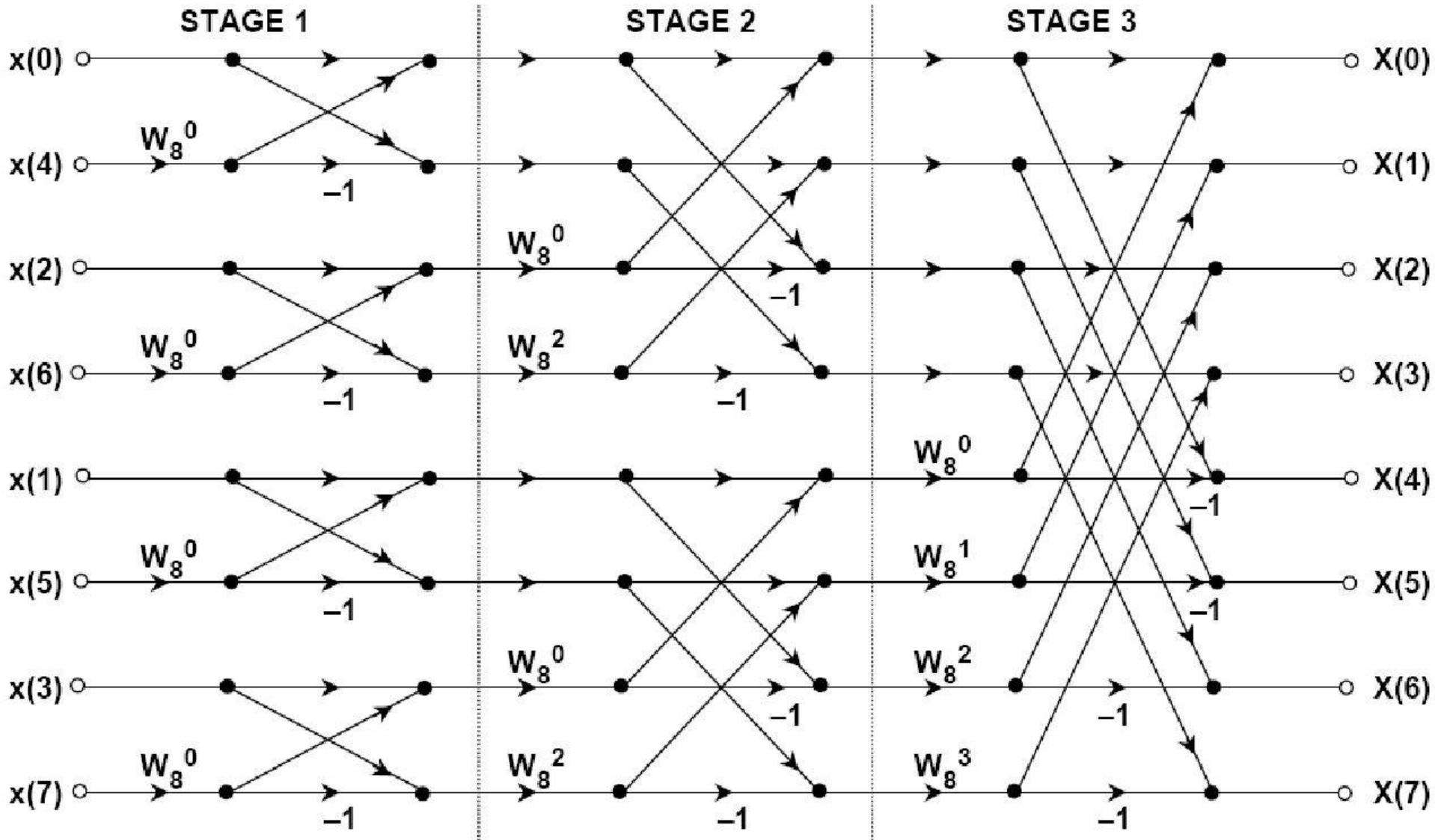
# **AIM:** Perform FFT and IFFT on discrete time signal.

# LAB EXP 5 DSIP

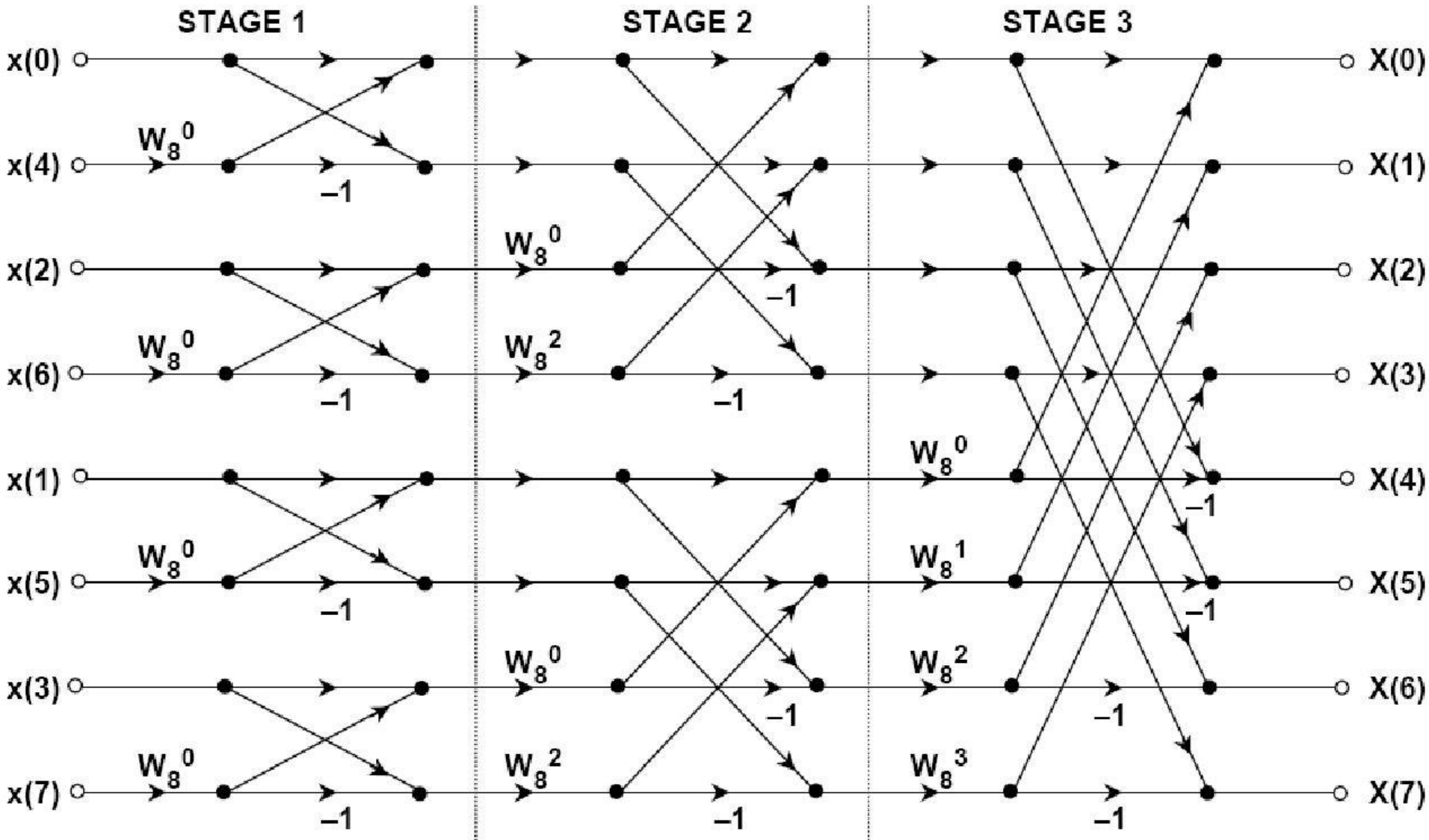
https://github.com/yugmakwana99/DSIP.git

NAME : YUG MAKWANA

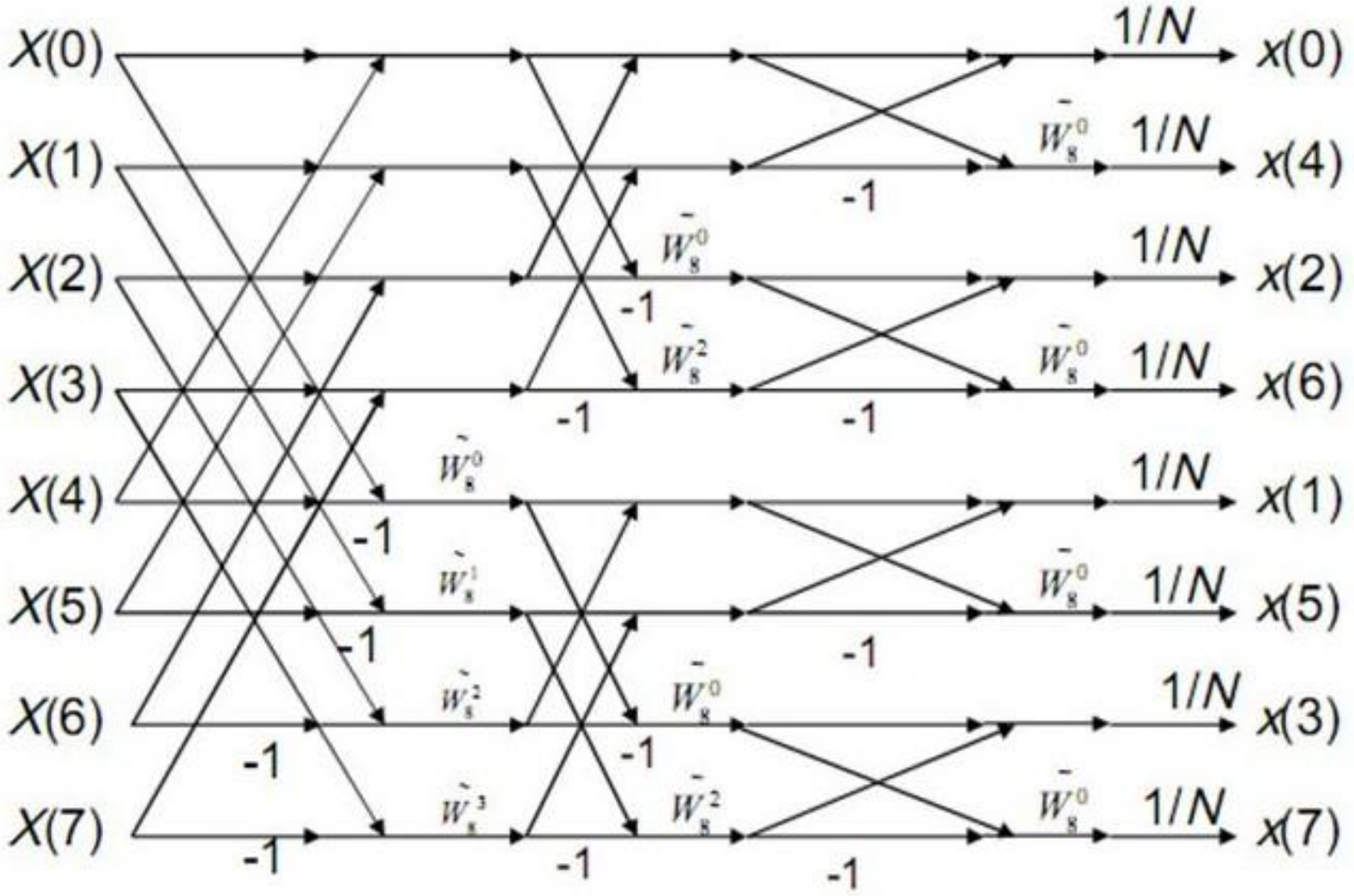
ENROLL : 92400133079


STAGE 1
STAGE 2
STAGE 3
x(0)
x(4)
x(2)
x(6)
x(1)
x(5)
x(3)
x(7)
X(0)
X(1)
X(2)
X(3)
X(4)
X(5)
X(6)
X(7)
W8 0
W8 2
W8 1
W8 3
–1

# DIT FFT


STAGE 1
STAGE 2
STAGE 3
x(0)
x(4)
x(2)
x(6)
x(1)
x(5)
x(3)
x(7)
X(0)
X(1)
X(2)
X(3)
X(4)
X(5)
X(6)
X(7)
$W_8^0$
$W_8^2$
$W_8^1$
$W_8^3$
–1

# Inverse FFT



# Program

```
import numpy as np
import matplotlib.pyplot as plt

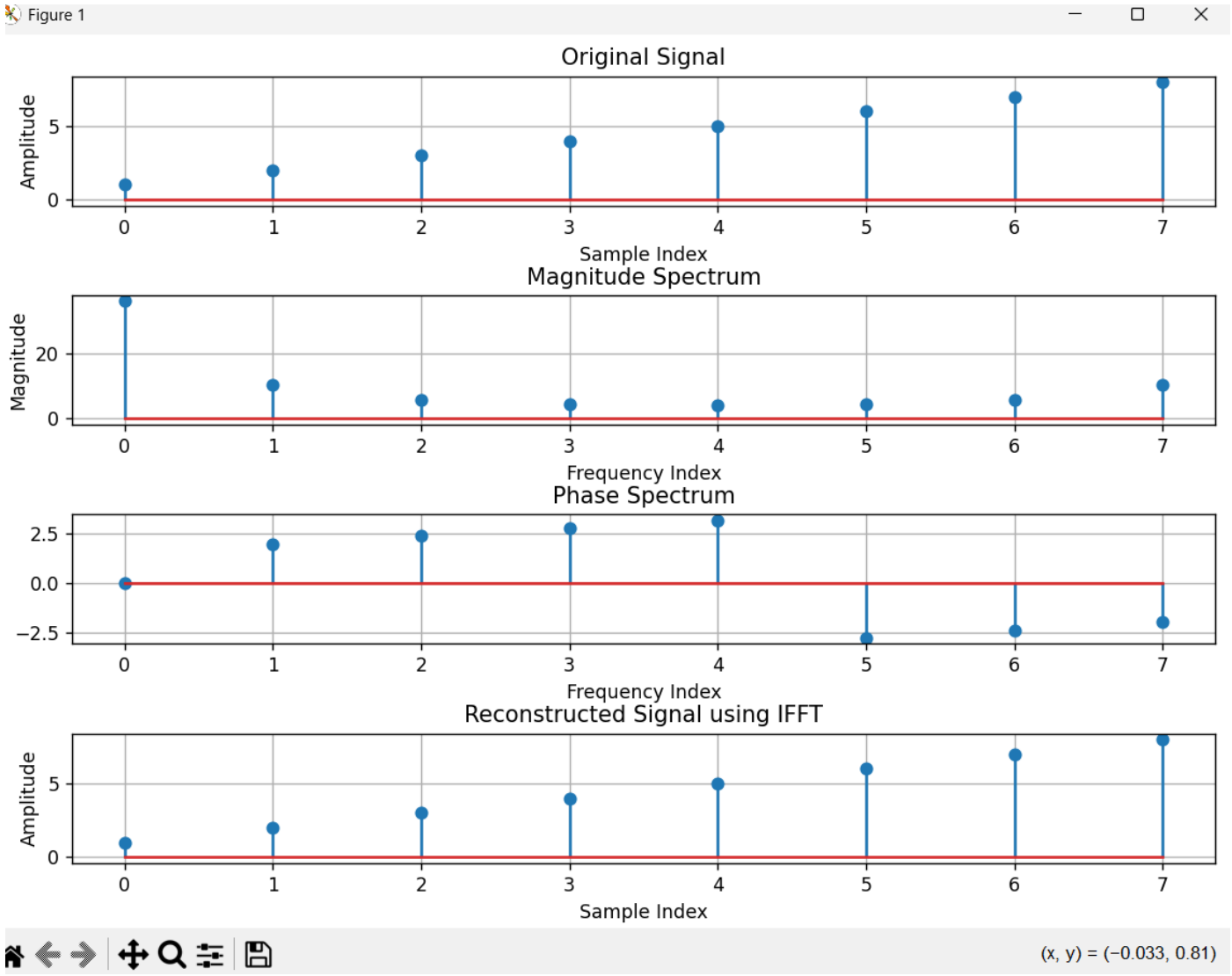
signal = np.array([1, 2, 3, 4, 5, 6, 7, 8])

# Compute the FFT of the signal
fft_result = np.fft.fft(signal)


# Compute the magnitude and Phase spectrum
of the FFT result
magnitude_spectrum = np.abs(fft_result)
Phase_spectrum = np.angle(fft_result)

# Compute the IFFT of the FFT result
reconstructed_signal =
np.fft.ifft(fft_result)
```

# OUTPUT


Figure 1
Original Signal
Amplitude
Sample Index
Magnitude Spectrum
Magnitude
Frequency Index
Phase Spectrum
Frequency Index
Reconstructed Signal using IFFT
Amplitude
Sample Index
(x, y) = (−0.033, 0.81)